KEY TERMS

Validation: a check that data entered is of the correct type and format; it does not guarantee that data is accurate

Verification: confirmation of data received by a system

Verification during data transfer

It is possible for data to be corrupted during transmission. Often, this happens when an individual bit is flipped from 1 to 0 or from 0 to 1. Verification techniques need to check on some property associated with the bit pattern.

The simplest approach is to use a simple one-bit parity check. This is particularly easy to do if data is transferred in bytes using a seven-bit code. Either even or odd parity can be implemented in the eighth bit of the byte. Assuming even parity, this is the procedure.

1. At the transmitting end, the number of 1s in the seven-bit code is counted.
2. If the count gives an even number, the parity bit is set to 0.
3. If the count gives an odd number, the parity bit is set to 1.
4. This is repeated for every byte in the transmission.
5. At the receiving end, the number of 1s in the eight-bit code is counted.
6. If the count gives an even number, the byte is accepted.
7. This is repeated for every byte in the transmission.

If no errors are found, the transmission is accepted. However, we cannot guarantee that the transmission is error free. It is possible for two bits to be flipped in an individual byte, which would mean that the transmission is incorrect but the parity check is passed. Fortunately, this is rather unlikely so it is sensible to assume no error. The limitation of the method is that it can only detect the presence of an error. It cannot identify the actual bit that is in error. If an error is detected, re-transmission has to be requested.

An alternative approach is to use the checksum method. At the transmitting end a block is defined as a number of bytes. Then, no matter what the bytes represent, the bits in each byte are interpreted as a binary number. The sum of these binary numbers in a block is calculated and supplied as a checksum value in the transmission. This is repeated for each block. The receiver does the same calculation and checks the sum of the numbers with the checksum value transmitted for each block in turn. Once again, an error can be detected but its position in the transmission cannot be determined.

TIP

Don't confuse a check digit with a checksum. A check digit is used for stored data; a checksum is only used for transmitted data.

Detecting the exact position of an error so as to correct it is considerably more complex. One approach is to use the parity block check method. Like the checksum method this is a longitudinal parity meaning that it is used to check a sequence of binary digits contained in a number of bytes.

WORKED EXAMPLE 9.01

Using a parity block check

At the transmitting end, a program reads a group of seven bytes as illustrated in Figure 9.01. The data is represented by seven bits for each byte. The most significant bit in each byte, bit 7, is undefined so we have left it blank.

	Seven-bit codes						
	1	0	1	0	0	1	1
	0	1	1	0	0	0	1
	1	0	1	1	0	0	0
	0	0	1	1	1	0	0
	0	1	1	0	0	1	0
	0	1	1	0	0	0	1
	0	1	1	0	0	0	1

Figure 9.01 Seven bytes to be transmitted

The parity bit is set for each of the bytes, as in Figure 9.02. The most significant bit is set to achieve even parity.

Parity bits	Seven-bit codes						
0	1	0	1	0	0	1	1
1	0	1	1	0	0	0	1
1	1	0	1	1	0	0	0
1	0	0	1	1	1	0	0
1	0	1	1	0	0	1	0
1	0	1	1	0	0	0	1
1	0	1	1	0	0	0	1

Figure 9.02 Bytes with the parity bit set

An additional byte is then created and each bit is set as a parity bit for the bits at that bit position. This includes counting the parity bits in the seven bytes containing data. This is illustrated in Figure 9.03.

Parity bits	Seven-bit codes							
0	1	0	1	0	0	1	1	
1	0	1	1	0	0	0	1	
1	1	0	1	1	0	0	0	
1	0	0	1	1	1	0	0	
1	0	1	1	0	0	1	0	
1	0	1	1	0	0	0	1	
1	0	1	1	0	0	0	1	
0	0	0	1	0	1	0	0	← **Parity byte**

Figure 9.03 Parity byte added

The program then transmits the eight bytes in sequence.

(Continued)

At the receiving end, a program takes the eight bytes as input and checks the parity sums for the individual bytes and for the bit positions.

Note that the method is handling a serial transmission so it includes longitudinal checking, but the checking algorithm is working on a matrix of bit values. If there is just one error in the seven bytes this method will allow the program at the receiving end to identify the position of the error. It can therefore correct the error so the transmission can be accepted.

Question 9.02

Assume that the seven bytes shown in Figure 9.04 contain data.

01001000	01000101	01110010	01100011

00101100	01010101	00110010

Figure 9.04 Seven bytes to be transmitted

The most significant bit is set to 0 but it is undefined at this stage because a seven-bit ASCII code represents character data. Choose a parity and change the value stored in the most significant bit to match this parity for each byte. Then create the eighth byte that would be used for transmission in a parity block check method.

Question 9.03

The eight bytes shown in Figure 9.05 have been received in a transmission using the parity block method. The first seven bytes contain the data and the last byte contains the parity check bits.

01001000	11000101	11110001	01100011
01001010	01010101	01110010	01110010

Figure 9.05 Eight bytes received in a transmission

a Identify what has gone wrong during the transmission.

b What would happen after the transmission is checked?

Reflection Point:

This chapter contains a lot of terminology. It is very easy to get confused about the definitions of the different terms used. Have you considered how you are going to attempt to remember all of the definitions and not get confused?

Summary

- Important considerations for the storage of data are: data integrity, data privacy and data security.
- Data protection laws relate to data privacy.
- Security measures for computer systems include authentication of users, prevention of unauthorised access, protection from malware and methods for recovery following system failure.
- Security methods for data include backup procedures, user authorisation and access control.
- Data entry to a system should be subject to data validation and data verification.
- Verification for data transmission may be carried out using: a parity check, a checksum or a parity block check method.

Exam-style Questions

1 a It is important that data has integrity.

i Identify the missing word in the sentence 'Concerns about the integrity of data are concerns about its [1]

ii Validation and verification are techniques that help to ensure data integrity when data is entered into a system.

Explain the difference between validation and verification. [3]

iii Define a type of validation and give an example. [2]

iv Even after validation has been correctly applied data may lack integrity when it comes to be used. Explain why that might happen. [2]

b Data should be protected from being read by unauthorised individuals.

Explain **two** policies that can be used to provide the protection. [4]

2 a Security of data is an important concern for a system administrator.

i Identify **three** reasons why data might not be available when a user needs it. [3]

ii Describe what could be features of a policy for ensuring data security. [3]

b It is important for mission-critical systems that there is a disaster recovery contingency plan in place.

i Define what type of disaster is under consideration here. [2]

ii Define what will be a major feature of the contingency plan. [2]

c Measures to ensure security of a computer system need to be in place on a daily basis if the system is connected to the Internet.

Describe **two** measures that could be taken to ensure security of the system. [4]

3 a When data is transmitted measures need to be applied to check whether the data has been transmitted correctly.

i If data consists of seven-bit codes transmitted in bytes, describe how a simple parity check system would be used. Your account should include a description of what happens at the transmitting end and what happens at the receiving end. [5]

ii An alternative approach is to use a checksum method. Describe how this works. [3]

b For either of these two methods there are limitations as to what can be achieved by them. Identify **two** of these limitations. [2]

c A different method which does not have all of these limitations is the parity block check method.

The following diagram represents eight bytes received where the parity block method has been applied at the transmitting end. The first seven bytes contain the data and the last byte contains parity bits.

Byte 1	0	1	0	1	0	0	1	1
Byte 2	1	0	1	1	0	0	0	1
Byte 3	1	1	0	1	0	0	0	0
Byte 4	1	0	0	1	1	1	0	0
Byte 5	1	0	1	1	0	0	1	0
Byte 6	1	0	1	1	0	0	0	1
Byte 7	1	0	1	1	0	0	0	1
Byte 8	0	0	0	1	0	1	0	0

Identify the problem with this received data and what would be done with it by the program used by the receiver. [4]

4 a Give the definition of the terms firewall and authentication. Explain how they can help with the security of data. [3]

b Describe **two** differences between data integrity and data security. [2]

c Data integrity is required at the input stage and also during transfer of the data.

i State **two** ways of maintaining data integrity at the input stage. Use examples to help explain your answer. [3]

ii State **two** ways of maintaining data integrity during data transmission. Use examples to help explain your answer. [3]

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5 Verification and validation can be applied during data entry.

Describe what is meant by these terms. For each method, explain why it is needed. [4]

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